Marco Vetrano

About Me

Computational Physicist and **Data Scientist**. Specialized in developing scalable **Machine Learning** models for insight extraction. I design and implement **AI** solutions to solve complex analytical problems.

Areas of Expertise

Physics • Machine Learning Network Analysis • Data Analysis

marco.vetrano01@gmail.com

Marco Vetrano

Marco Vetrano01

Experience

2023–Present — **Quantum Machine Learning R&D**
University of Palermo · Italy

- Development of hybrid (Quantum-Classical) algorithms for analyzing and classifying large astrophysical datasets (PNRR STILES Project). Code deployment on Leonardo HPC and IBM Quantum devices.
- **Stack**: Python, Linux, Scikit-Learn, PyTorch.

2025 — **Quantum Machine Learning R&D**
IFISC - University of the Balearic Islands · Spain

- Design and implementation of predictive models for chaotic time series analysis. Code deployment on IFISC-owned HPC cluster.
- **Stack**: Python, Linux, Docker, Scikit-Learn, PyTorch.

Education

2026 — **PhD in Physics**
· University of Palermo

2026 - Present — **Master in AI Development and Engineering**
ProfessionAI ·

2023 — **Master's Degree in Physics**
· University of Palermo

2019 — **Bachelor's Degree in Physical Sciences**
· University of Palermo

Languages

Italian	**C2**	Native Speaker	
English	**C1**	**Writing** C1	**Reading** C1
		Listening C1	**Speaking** C1

Tech Stack

Python · SQL · C++ · Docker · Pytorch · Hugging Face · TensorFlow · XGBoost · Langchain · Git · FastAPI · Linux · AWS · Quarto · LaTeX · PySpark

Publications

2025 "State estimation with quantum extreme learning machines beyond the scrambling time.", Springer Nature npj Quantum Information 11.1 (2025): 20.

2025 "Exoplanetary atmospheres retrieval via a quantum extreme learning machine.", arXiv preprint arXiv:2509.03617.

Certifications

2026 Udemy: Machine Learning and Data Science in Python

2023 Duolingo English Test - C1 Certified

Projects

2025	**QuantumToolBox** University of Palermo · Italy A Python package developed using numpy, scikit-learn, and plotly for simulating quantum dynamics and implementing classification and time-series prediction models. Available on GitHub in the QuantumStuff repository.
2025-2026	**AstroQELM** University of Palermo · Italy An end-to-end Python package for spectral data analysis. Features include h5/txt parsing, pre-processing/compression via scikit-learn, and processing/classification using qiskit, numpy, and scikit-learn. Available in the AstroQELM repository.
2026	**Instagram-Analyzer** · Italy Backend application developed with FastAPI for automated Instagram scraping. It extracts data from user profiles (reels, comments) using Apify actors, calculating metrics like BPM. Includes Hugging Face models for Speech-to-Text and sentiment analysis. Fully containerized.
2026	**Olist-App** · Italy End-to-end analysis of the Olist e-commerce dataset (Kaggle). Containerized with Docker, it uses SQLAlchemy and PostgreSQL for data management. Features three progressive sentiment analysis models (Scikit-Learn, PyTorch, Hugging Face). Ongoing development includes a Streamlit UI and a custom LLM agent.

Other Experience

2022 & 2024 — **Academic Tutor (STEM)**
University of Palermo · Italy

- Teaching support for Physics, Mathematical Analysis, and Geometry courses.
- Personalized mentoring for students with learning disabilities (DSA).

2022 — **Data Analyst Intern**
Ri.MED Foundation · Italy
Applied Network Analysis techniques to study RNA structures, identifying critical interaction patterns.

2018 — **Data Analyst Intern**
IASF - Institute of Space Astrophysics and Cosmic Physics · Italy
Developed C scripts for spectral processing and analysis of black hole data.

Awards

2023 First Place - **QuHack4AI: Quantum Hackathon for Industrial Applications** (University of Bologna & Unipol). Project: Variational algorithms for hydrogen and water molecules.

2023 **"Eduardo Gugino" Award** for Master's Thesis: "Information Scrambling in Quantum Extreme Learning Machines".

Talks

2023 "Information Scrambling in Quantum Extreme Learning Machines for State Estimation" at New Trends in Non-equilibrium Statistical Mechanics

2024 "QELM response to scrambling dynamics and applications to atmospheric retrieval" at International School on Non-Equilibrium Phenomena

2025 "Exoplanetary Retrieval with Quantum Extreme Learning Machines" at the National Congress of the Italian Physical Society

Marco Vetrano marco.vetrano01@gmail.com Palermo, Italy +39 3459460312
https://marcovetrano01.github.io/Portfolio/
I authorize the processing of my personal data contained in this CV in accordance with local laws and the GDPR (EU Regulation 2016/679).